

Embedded Linux & System Software Interview Handbook

Phase 10 – Virtualization, Containers & Cloud-Native Linux

This phase introduces virtualization technologies, Linux containers, orchestration, and cloud-native concepts that are increasingly used in modern embedded, edge, and server platforms.

Chapter 17 – Virtualization, Containers & Cloud-Native Linux

Topics Covered

- Virtualization Fundamentals
- Type-1 vs Type-2 Hypervisors
- Hardware Virtualization (Intel VT-x, AMD-V, ARM Virtualization)
- KVM Architecture
- QEMU
- libvirt
- Virtual Machine Lifecycle
- Memory Virtualization
- CPU Virtualization
- I/O Virtualization
- SR-IOV
- VFIO and PCI Passthrough
- Linux Namespaces
- cgroups
- Container Runtime Architecture
- Docker
- containerd
- CRI-O
- OCI Standards
- Kubernetes Architecture
- Pods, Deployments and Services
- Persistent Volumes
- Networking in Kubernetes
- Helm
- Service Mesh Overview
- Edge Computing
- Confidential Computing
- Kata Containers

- MicroVMs (Firecracker)
- Cloud-Native Observability
- Production Deployment Patterns
- Senior Interview Questions

Learning Outcomes

After completing this phase, you will understand virtualization and container technologies, deploy workloads using Kubernetes, work with cloud-native Linux platforms, and build secure, scalable systems for embedded, edge, and cloud environments.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Phase 5	Interrupts, Timers & Deferred Work
Phase 6	Advanced Device Drivers
Phase 7	Linux Security & Secure Boot
Phase 8	Performance & Kernel Debugging
Phase 9	Embedded Linux Build Systems & BSP
Phase 10	Virtualization, Containers & Cloud-Native Linux
Estimated Size	250–350 pages when expanded
Target	Senior Linux / Cloud / Platform Engineering Interviews